Lecture 26

Bayesian Hierarchical models

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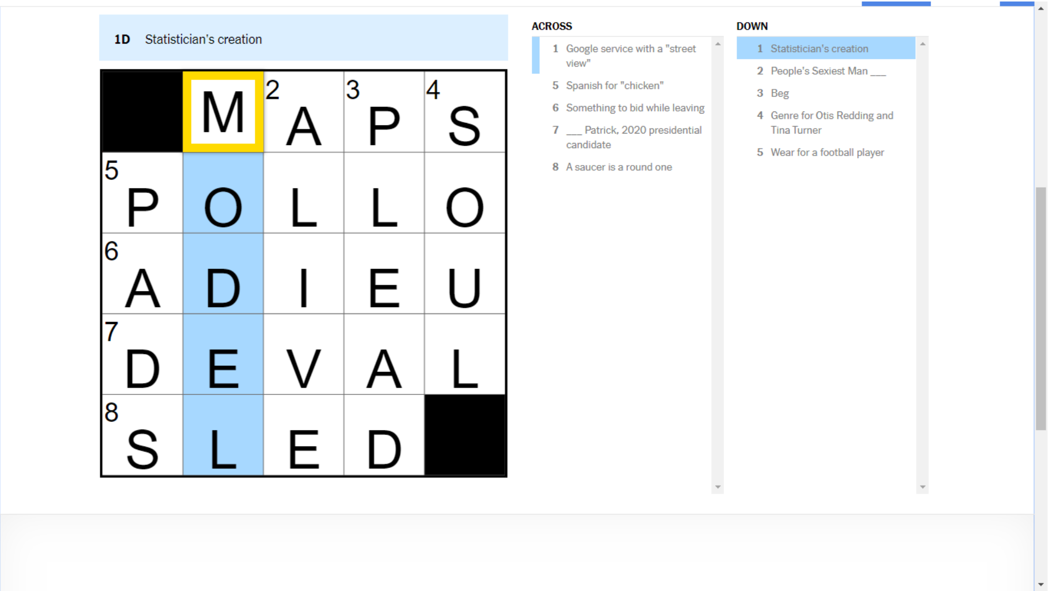
(Edited from the lecture notes by Brian J. Reich, NC State)

Hierarchical models

- Hierarchical modeling provides a framework for building complex and high-dimensional models from simple and low-dimensional building blocks.
- Of course, it is possible to analyze these models using non-Bayesian methods.
- However, this modeling framework is popular in the Bayesian literature because MCMC is conducive to hierarchical models.
- Both “divide and conquer” big problems by splitting them into a series of smaller problems in the same way.

We build models!



Outline oc Chapter 6

- Building a hierarchical model through layers
- Directed acyclic graphs
- Several examples

Hierarchical models

Often Bayesian models can we written in the following layers of the hierarchy

1. **Data layer**: $[\mathbf{Y}|\theta, \alpha]$ is the likelihood for the observed data **Y** given the model parameters.

2. **Process layer**: $[\theta|\alpha]$ is the model for the parameters $\boldsymbol{\theta}$ that define the latent data generating process.

3. **Prior layer**: $[\alpha]$ prior for hyperparameters.

Epidemiology example - Data layer

- Let S_t and I_t be the number of susceptible and infected individuals in a population, respectively, at time t.
- The data Y_t is the number of observed cases at time t.
- The data layer models our ability to measure the process I_t.
- **Data layer**: $Y_t | I_t \sim \text{Binomial}(I_t, p)$.
- This assumes no false positives and false negative probability p.

Epidemiology example - Process layer

- Scientific understanding of the disease is used to model disease propagation.
- We might select the simple Reed-Frost model

$$\textbf{Process layer:} \qquad I_{t+1} \sim \text{Binomial}\left[S_t, 1-(1-q)^{I_t}\right],$$
$$S_{t+1} = S_t - I_{t+1}.$$

- This assumes all infected individuals are removed from the population before the next time step.
- Also that q is the probability of a non-infected person coming into contact with and contracting the disease from an infected individual.

Epidemiology example - Prior layer

- The epidemiological process-layer model expresses the disease dynamics up to a few unknown parameters.
- The Bayesian model is completed using priors, say,
- **Prior layer:**

$$
\begin{aligned}
I_1 &\sim \text{Poisson}(\lambda_1), \\
S_1 &\sim \text{Poisson}(\lambda_2), \\
p, q &\sim \text{beta}(a, b).
\end{aligned}
$$

When to stop adding layers?

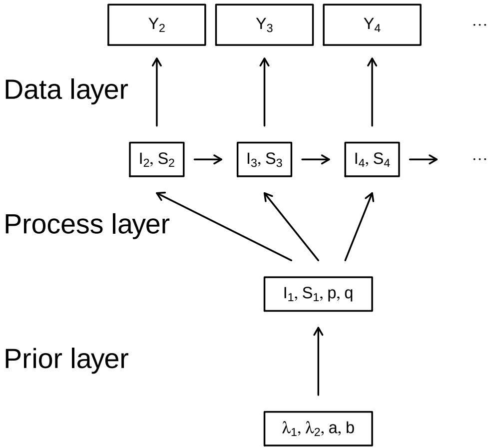
- In the previous example a, b, λ_1 and λ_2 are fixed.
- But we will have uncertainty about the correct value.
- Maybe replace a fixed value with another layer, say $a \sim \text{Uniform}(0, \theta)$?
- Then maybe $\theta \sim \text{Exponential}(\xi)$, $\xi \sim \text{Uniform}(0, \eta)$, etc.
- Rule of thumb: Be careful assigning priors to parameters in layers without replication.
- For example, even if we knew p exactly this would be just one value and we couldn't hope to estimate the parameters of its beta distribution.

Directed acyclic graphs (DAGs)

- A DAG is a graphical representation of a hierarchical model.
- DAGS sometimes go by the name Bayesian networks.
- Each observation and parameter is a node.
- An arrow for X to Y means that the conditional distribution of Y depends on X.
- "Directed" means that arrows only go one way.
- Acyclic means there are no cycles, e.g.,

$$X \to Y \to Z \to X.$$

Epidemiology example - DAG



Directed acyclic graphs (DAGs)

- Building models this way ensures we will always have a valid joint distribution.
- For example, say we need to specify the joint distribution of (X, Y, Z).
- Any joint distribution can be written as

$$f(X, Y, Z) = f(X)f(Y|X)f(Z|X, Y).$$

- This is a fully-connected DAG.
- Ad-hoc constructions like

$$f(X, Y, Z) = f(X|Z)f(Y|X)f(Z|X, Y)$$

may or may not give a valid joint PDF.

Hierarchical models and MCMC

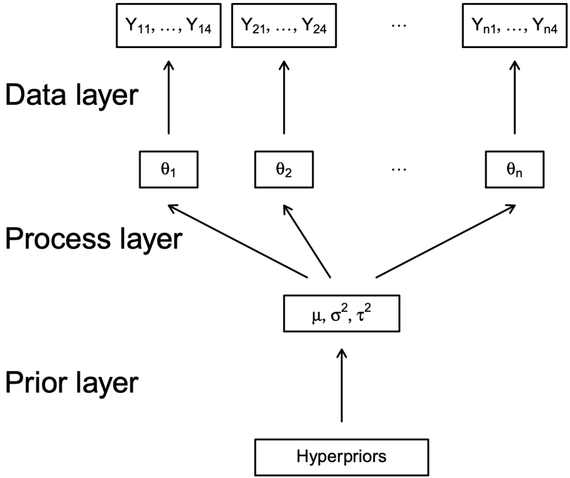
- Consider the classic one-way random effects model:

$$Y_{ij} \sim \mathrm{N}(\theta_i, \sigma^2) \quad \text{and} \quad \theta_i \sim N(\mu, \tau^2)$$

  where Y_{ij} is the j^{th} replicate for unit i and $\boldsymbol{\alpha} = (\mu, \sigma^2, \tau^2)$ has an uninformative prior.

- This hierarchy can be written using a directed acyclic graph.

Random effects example - DAG



Hierarchical models and MCMC

- MCMC is efficient in this case even if the number of parameter or levels of the hierarchy is large.
- You only need to consider “connected nodes” when you update each parameter.
- 1. $[\theta_i|\cdot]$
  2. $[\mu|\cdot]$
  3. $[\sigma^2|\cdot]$
  4. $[\tau^2|\cdot]$
- Each of these updates is a draw from a standard one-dimensional normal or inverse gamma.

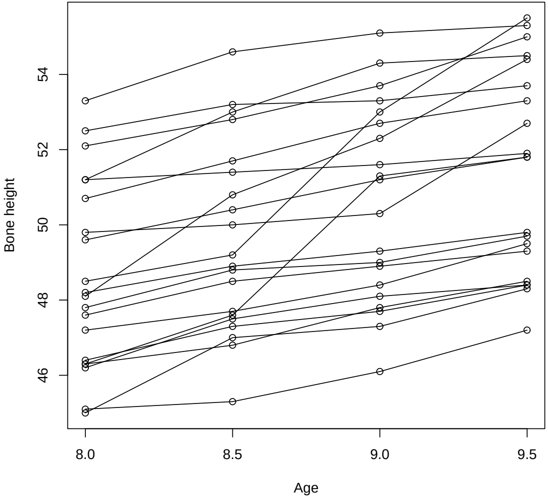
Random slopes model

- Let Y_{ij} be the j^{th} observation for subject i.
- As an example, consider the data plotted on the next slide were Y_{ij} is the bone density for child i at age X_j.
- Here we might specify a different regression for each child to capture variability over the population of children:

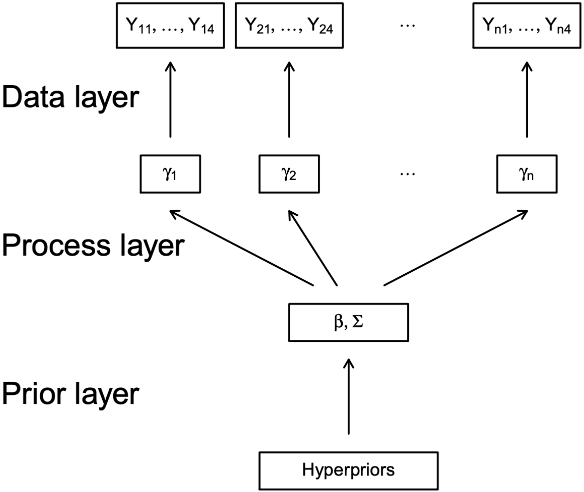
$$Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_j\gamma_{1i}, \sigma^2).$$

- $\gamma_i = (\gamma_{i0}, \gamma_{i1})^T$ controls the growth curve for child i.
- These separate regression are tied together in the prior, $\gamma_i \sim \text{Normal}(\beta, \Sigma)$, which borrows strength across children.
- This is a linear mixed model: γ_i are random effects specific to one child and β are fixed effects common to all children.

Bone height data



Draw a DAG and work out the full conditionals



Thank you!